

The Projective-Orbit Retry Theorem, Expanded

A detailed proof of Theorem 7.1 and a complete calculation of the unit-square distance matrix

Quantum Hodology Research Project

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Abstract

Theorem 7.1 of *Exact Low-Dimensional Hodology* gives a simple but powerful recipe: encode a group orbit in a quantum system, make a displacement of metric length r succeed with probability $1/r$, and charge one unit for every attempt. Retrying then costs r on average, and the triangle inequality proves that no adaptive sequence of other actions is cheaper. This note explains every assumption and every inequality in that argument. It also corrects a left-versus-right invariance convention that is harmless for the abelian examples in the paper but matters for nonabelian groups. Finally, it derives every entry of the four-by-four matrix D in Eq. (43), checks the Bellman equations directly, and performs the complete Schoenberg calculation showing that D is exactly the Euclidean metric of a unit square. Finally, it removes the assumption that the agent is handed or updates a group label: the controller receives only its raw action–outcome history, and SIC-outcome goals retain the same square geometry after subtraction of a derived common reporting overhead. The expanded control analysis makes the reward convention explicit, evaluates all four actions in every predictive history class, and solves the discounted problem in closed form. It shows that the square is specifically an undiscounted expected-hitting-time geometry: discounting can change the optimal policy and collapse its spatial distinctions. The finite Bellman calculation is justified by proving that future SIC statistics induce four causal equivalence classes of histories, not by supplying a hidden binary location variable.

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1 What the theorem is trying to accomplish

Suppose an agent has operational situations indexed by elements x of a group G . An action indexed by $h \in G$ attempts to translate the situation. The desired difficulty of displacement h is a prescribed group metric length

$$r_h = d_G(\mathbf{e}, h).$$

The construction turns this length into a waiting time. One attempt costs one intervention and succeeds with probability

$$p_h = \frac{1}{r_h}.$$

If failure changes nothing, repeated attempts have geometric mean waiting time $1/p_h = r_h$. This immediately supplies a policy whose expected cost equals the desired distance. The nontrivial half of the theorem is to prove that a clever combination of shorter displacements, stochastic choices, or history-dependent actions cannot do better.

The proof uses the metric distance to the goal as a *potential*:

$$V_g(x) = d_G(x, g).$$

The triangle inequality bounds how much any successful action can reduce this potential. The probability $p_h = 1/r_h$ is calibrated so that the *expected* decrease of V_g in one unit-cost intervention is at most one. Reducing an initial potential $V_g(x)$ to zero therefore costs at least $V_g(x)$ on average. Direct retry attains this bound.

2 The assumptions, one by one

2.1 A group of displacements

A group G has an identity $\mathbf{e}$, inverses, and an associative product. The product records composition of displacements. For example, in the square construction

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2,$$

with componentwise addition modulo two. Its elements are 00, 10, 01, 11. Every nonidentity element is its own inverse.

2.2 A translation-invariant metric

A metric d_G is *left invariant* if

$$d_G(ax, ay) = d_G(x, y) \quad \text{for every } a, x, y \in G,$$

and *right invariant* if

$$d_G(xa, ya) = d_G(x, y).$$

It is *bi-invariant* if it has both properties. On an abelian group the left and right translations coincide, so either invariance implies the other.

The normalization

$$\min_{h \neq \mathbf{e}} d_G(\mathbf{e}, h) = 1$$

is not cosmetic. It guarantees $r_h \geq 1$, hence $0 < p_h = 1/r_h \leq 1$, as required for a probability. It also fixes the unit of cost: the shortest nontrivial displacement costs one expected intervention.

2.3 A projective quantum orbit

A projective unitary representation assigns a unitary U_g to every $g \in G$, with multiplication respected up to global phase:

$$U_g U_h = e^{i\phi(g,h)} U_{gh}.$$

Global phase disappears under conjugation, so the density operators

$$\rho_g = U_g \rho_0 U_g^\dagger$$

carry an exact group action. The distinct-orbit assumption is

$$\rho_g = \rho_h \implies g = h.$$

It says that two different group labels do not denote exactly the same physical density operator. It does *not* say that the states are orthogonal or perfectly distinguishable in one shot.

The cost proof itself is a statement about the induced controlled group process. Assumption (2.3) supplies its physical quantum realization and prevents an exact stabilizer from silently identifying two purported places.

2.4 The retry instrument

For every nonidentity displacement h , set

$$K_s^{(h)} = \sqrt{p_h} U_h, \quad K_f^{(h)} = \sqrt{1 - p_h} I, \quad p_h = \frac{1}{d_G(\mathbf{e}, h)}. \quad (1)$$

The labels s and f mean success and failure. This is a valid two-outcome quantum instrument because

$$K_s^{(h)\dagger} K_s^{(h)} + K_f^{(h)\dagger} K_f^{(h)} = p_h U_h^\dagger U_h + (1 - p_h) I \quad (2)$$

$$= p_h I + (1 - p_h) I = I. \quad (3)$$

Conditional on failure, normalization removes the scalar and the state is unchanged. Conditional on success, normalization removes $\sqrt{p_h}$ and the state is conjugated by U_h .

3 A convention issue in the printed statement

The printed theorem lets success act by left multiplication, $x \mapsto hx$, but assumes only that d_G is left invariant. It then uses

$$d_G(x, hx) = d_G(\mathbf{e}, h).$$

For a general nonabelian group, that equality follows from *right* invariance, not left invariance: right multiplication of both arguments by x^{-1} sends (x, hx) to $(\mathbf{e}, h)$.

There are two equally valid repairs.

1. Keep the printed dynamics $x \mapsto hx$ and assume that d_G is right invariant. The direct displacement is $h = gx^{-1}$.
2. Keep a left-invariant metric but use right-action dynamics $x \mapsto xh$. The direct displacement is $h = x^{-1}g$.

The paper's exact examples use abelian groups such as $\mathbb{Z}_2^2$, $\mathbb{Z}^2$, and finite phase tori. Their metrics are bi-invariant, so the displayed proof and every reported result are unchanged. The distinction matters only when generalizing the theorem to nonabelian action groups.

For definiteness, the next section gives the corrected left-action/right-invariant form. Reversing every product gives the left-invariant/right-action form verbatim.

4 The theorem with a fully explicit proof

Theorem 4.1 (Projective-orbit retry construction, convention-correct form). *Let G be finite or countable and let d_G be a right-invariant metric normalized by (2.2). Suppose G has a projective unitary representation and a fiducial density operator satisfying (2.3). Under action $h \neq \mathbf{e}$, failure leaves x fixed and success sends x to hx , with the instrument (1). Every attempt costs one. If the goal is g , then the optimal expected hitting cost is*

$$D(x, g) = d_G(x, g).$$

Proof. We separate the proof into validity, achievability, and optimality.

1. The quantum actions are valid. Equation (3) proves trace preservation after summing over the two reported branches. Because $d_G(\mathbf{e}, h) \geq 1$, both coefficients are real and lie between zero and one.

2. Direct retry achieves the metric distance. Assume $x \neq g$ and choose

$$h_* = gx^{-1}.$$

On success, $h_*x = g$, so one successful event reaches the goal. On failure, the state and group coordinate remain at x , so the same experiment starts again from exactly the same situation. The number T of attempts until the first success therefore has

$$\Pr(T = t) = (1 - p_{h_*})^{t-1} p_{h_*}, \quad t = 1, 2, \dots$$

This is the geometric distribution, whose mean is

$$\mathbb{E}T = \frac{1}{p_{h_*}} = d_G(\mathbf{e}, h_*).$$

Right invariance, applied by multiplying both arguments by x , gives

$$d_G(\mathbf{e}, gx^{-1}) = d_G(x, g).$$

Thus direct retry has expected cost $d_G(x, g)$, proving

$$D(x, g) \leq d_G(x, g). \tag{4}$$

3. No single action can beat the distance potential. Define

$$V(y) = d_G(y, g).$$

If the current point is y and action h is selected, the expected one-step cost plus future potential is

$$Q_h(y) = 1 + (1 - p_h)V(y) + p_hV(hy). \tag{5}$$

The reverse triangle inequality follows from the ordinary triangle inequality:

$$V(y) = d_G(y, g) \leq d_G(y, hy) + d_G(hy, g) = d_G(y, hy) + V(hy).$$

Consequently

$$V(y) - V(hy) \leq d_G(y, hy).$$

Right invariance gives

$$d_G(y, hy) = d_G(\mathbf{e}, h) = \frac{1}{p_h},$$

and hence

$$p_h[V(y) - V(hy)] \leq 1. \tag{6}$$

Rearranging (5),

$$Q_h(y) = V(y) + 1 - p_h[V(y) - V(hy)] \geq V(y). \tag{7}$$

Thus every allowed action has Bellman cost at least the metric potential.

4. Why this also rules out adaptive policies and mixtures. Conditioned on the complete past, a policy may choose any action, possibly at random. Inequality (6) says that the conditional expected decrease of V during that intervention is at most one for every pure choice. Averaging preserves the inequality for a randomized choice. After n interventions, the expected total decrease is therefore at most n . For a policy with hitting time τ , apply this statement to $\tau \wedge n$ and let $n \rightarrow \infty$. At the goal, $V = 0$, so every proper policy of finite expected duration obeys

$$V(x) \leq \mathbb{E}\tau.$$

Policies with infinite expected duration cannot improve the optimum. Hence

$$D(x, g) \geq V(x) = d_G(x, g). \quad (8)$$

Combining (4) and (8) proves equality. $\square$

4.1 Where equality occurs

For the direct action $h_* = gx^{-1}$, success lands at g , so $V(h_*x) = 0$, while $1/p_{h_*} = V(x)$. Therefore

$$Q_{h_*}(x) = 1 + \left(1 - \frac{1}{V(x)}\right) V(x) = V(x).$$

All alternative actions satisfy $Q_h(x) \geq V(x)$; some may tie, but none can be smaller. This is the precise sense in which the triangle inequality is the optimality certificate.

5 Eq. (43): the unit-square example

5.1 The group and its desired metric

Take

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{00, 10, 01, 11\}.$$

Addition is componentwise modulo two. We associate these labels with the four Euclidean points

$$00 \leftrightarrow (0, 0), \quad 10 \leftrightarrow (1, 0), \quad 01 \leftrightarrow (0, 1), \quad 11 \leftrightarrow (1, 1). \quad (9)$$

Define the translation-invariant group metric by its displacement lengths:

$$d_G(\mathbf{e}, 10) = 1, \quad d_G(\mathbf{e}, 01) = 1, \quad d_G(\mathbf{e}, 11) = \sqrt{2}, \quad (10)$$

and

$$d_G(x, g) = d_G(\mathbf{e}, g - x).$$

Here subtraction equals addition because every element has order two. The only nontrivial triangle inequality is $\sqrt{2} \leq 1 + 1$; hence this is indeed a metric. Since G is abelian, it is both left and right invariant.

5.2 The qubit orbit

The physical system is a qubit with fiducial Bloch vector

$$\mathbf{r}_{00} = \frac{1}{\sqrt{3}}(1, 1, 1).$$

Conjugation by Pauli X and Z generates four states:

$$\begin{aligned} \mathbf{r}_{00} &= (+, +, +)/\sqrt{3}, & \mathbf{r}_{10} &= (+, -, -)/\sqrt{3}, \\ \mathbf{r}_{01} &= (-, -, +)/\sqrt{3}, & \mathbf{r}_{11} &= (-, +, -)/\sqrt{3}. \end{aligned}$$

They are distinct and form a tetrahedron on the Bloch sphere. Every distinct pair has overlap

$$\text{Tr}(\rho_x \rho_g) = \frac{1}{3}.$$

Thus their *physical-state geometry* is equidistant. The square below is instead the geometry of optimal controlled cost.

5.3 Removing the hidden-location assumption with the SIC instrument

The labels 00, 10, 01, 11 should not be supplied as an unobserved location register. The tetrahedral orbit itself defines a four-outcome qubit SIC. Let $\Pi_o = \rho_o$ be its rank-one projectors and define

$$M_o = \frac{1}{\sqrt{2}}\Pi_o, \quad o \in \{00, 10, 01, 11\}. \quad (11)$$

Because a qubit SIC obeys $\sum_o \Pi_o = 2I$,

$$\sum_o M_o^\dagger M_o = \frac{1}{2} \sum_o \Pi_o = I.$$

Thus $\{M_o\}$ is a valid four-outcome instrument.

If its input is orbit state ρ_x , then

$$p(o | x) = \text{Tr}(M_o \rho_x M_o^\dagger) = \frac{1}{2} \text{Tr}(\Pi_o \Pi_x) = \begin{cases} 1/2, & o = x, \\ 1/6, & o \neq x. \end{cases} \quad (12)$$

The outcome does not reveal the *premeasurement* orbit label perfectly; four nonorthogonal qubit states cannot be perfectly discriminated. The key fact is instead the conditional update

$$\frac{M_o \rho_x M_o^\dagger}{p(o | x)} = \Pi_o. \quad (13)$$

Every possible outcome o prepares the corresponding tetrahedral state. Immediately after the SIC action, the agent therefore knows its current operational state: it is the last observed outcome token. There is no hidden coordinate inference in this statement. The binary strings are convenient theorist names and may be replaced by arbitrary colors or integers.

The agent can learn what opaque movement buttons do as well. Starting from an observed SIC anchor o , it presses a button and performs the same SIC again. The next-outcome law has its $1/2$ peak at the translated anchor and $1/6$ elsewhere. Repeating this common test recovers the induced permutations of the four outcome tokens, up to the unavoidable choice of origin, axes, and signs. Thus both “where I am” and “what this action does” can be grounded in observed statistics rather than privileged binary coordinates.

5.4 The three displacement actions

The axial displacements 10 and 01 have length one, so their retry probabilities are one. They are deterministic Pauli channels:

$$10 : \rho \mapsto X\rho X, \quad 01 : \rho \mapsto Z\rho Z.$$

The diagonal displacement 11 is implemented by Y , projectively equivalent to XZ , and has

$$p_{11} = \frac{1}{\sqrt{2}}.$$

Therefore

$$K_s = \sqrt{p_{11}}Y = 2^{-1/4}Y, \quad K_f = \sqrt{1 - \frac{1}{\sqrt{2}}}I. \quad (14)$$

Repeated diagonal attempts have expected cost

$$\frac{1}{p_{11}} = \sqrt{2}.$$

Taking two deterministic axial steps would cost 2, so direct diagonal retry is better. The theorem proves that no adaptive alternative costs less than $\sqrt{2}$.

5.5 Computing every entry of D

Order rows and columns as 00, 10, 01, 11. The diagonal entries vanish because the agent is already at its goal:

$$D(00, 00) = D(10, 10) = D(01, 01) = D(11, 11) = 0.$$

Pairs differing in exactly one bit are axial neighbors and cost one:

$$\begin{aligned} D(00, 10) &= 1, & D(00, 01) &= 1, \\ D(10, 11) &= 1, & D(01, 11) &= 1. \end{aligned}$$

Pairs differing in both bits require displacement 11 and cost $\sqrt{2}$:

$$D(00, 11) = \sqrt{2}, \quad D(10, 01) = \sqrt{2}.$$

The metric is symmetric, so reversing any pair gives the same entry. Hence

$$D = \begin{pmatrix} 0 & 1 & 1 & \sqrt{2} \\ 1 & 0 & \sqrt{2} & 1 \\ 1 & \sqrt{2} & 0 & 1 \\ \sqrt{2} & 1 & 1 & 0 \end{pmatrix}. \quad (15)$$

The same calculation can be displayed as a displacement table:

source	goal	group difference	optimal expected cost
00	10	10	1
00	01	01	1
00	11	11	$\sqrt{2}$
10	01	11	$\sqrt{2}$
10	11	01	1
01	11	10	1

6 A direct Bellman check for the square

It is useful to verify one column of D without invoking the theorem as a black box. Choose goal $g = 11$. From (15), the proposed values are

$$V(00) = \sqrt{2}, \quad V(10) = 1, \quad V(01) = 1, \quad V(11) = 0.$$

At 00, deterministic X or Z gives

$$Q_X(00) = 1 + V(10) = 2, \quad Q_Z(00) = 1 + V(01) = 2.$$

The diagonal action succeeds at 11 with probability $1/\sqrt{2}$ and otherwise remains at 00, so

$$\begin{aligned} Q_Y(00) &= 1 + \left(1 - \frac{1}{\sqrt{2}}\right) V(00) + \frac{1}{\sqrt{2}} V(11) \\ &= 1 + \left(1 - \frac{1}{\sqrt{2}}\right) \sqrt{2} \\ &= 1 + \sqrt{2} - 1 = \sqrt{2}. \end{aligned}$$

Thus the diagonal action attains the proposed value and the axial alternatives are more expensive.

At 10, the deterministic Z action reaches 11 in one step, so $V(10) = 1$. Similarly, X sends 01 to 11 in one step. At the goal, the process terminates with value zero. The other goal columns follow by translation symmetry. This explicitly verifies the stochastic-shortest-path Bellman equations whose solution is (15).

7 The complete history-only reinforcement-learning specification

The phrase “one-step cost” can obscure several choices that must be made to define a reinforcement-learning problem. This section spells out the entire model without giving the agent a location variable: what it observes, what counts as one time step, when an episode ends, what reward is paid, and why an exact calculation need not enumerate infinitely many histories.

7.1 The agent receives only an action–outcome history

A *decision epoch* occurs whenever the agent chooses one quantum instrument. One use of the X channel, one use of the Z channel, one attempt of the random- Y instrument, and one use of the SIC instrument each count as exactly one step. A step is therefore an intervention, not a successful displacement and not an elapsed laboratory time chosen after the outcome is known.

The controller is *not* supplied a variable called “current position.” Its information at time t is the literal record

$$h_t = (a_0, o_1, a_1, o_2, \dots, a_{t-1}, o_t),$$

consisting only of its chosen actions and their observed outcomes. A policy is therefore a map $\pi(a \mid h_t, g)$, where g is the requested SIC outcome token. The four SIC outcomes form an arbitrary alphabet $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$. They need not be called 00, 01, 10, 11, and the agent never updates such a binary label after pressing a movement button.

The episode terminates when the agent chooses the SIC instrument and observes the requested token g . Merely having observed g earlier does not end a new episode. Thus the goal itself is an observable event, not occupation of a theorist-supplied state.

7.2 The four instruments and their observable alphabets

The action set is

$$\mathcal{A} = \{X, Z, \tilde{Y}, S\},$$

where $\tilde{Y}$ denotes the two-outcome retry instrument and S the four-outcome SIC instrument. The data available to the agent on one use are:

chosen instrument	observed outcome
X	one acknowledgement symbol $\bullet_X$
Z	one acknowledgement symbol $\bullet_Z$
$\tilde{Y}$	success s or failure f , with success probability $p = 1/\sqrt{2}$
S	one token $o \in \Omega$

In particular, the acknowledgement from X or Z does not say where the system went. The controller may condition its next action on the string $(S, o), (X, \bullet_X)$, but it is not told to replace o by some other location label. The physical consequences of a button are accessible only through distributions of later outcomes, especially a later SIC test.

7.3 Reward, cost, stopping time, and the absence of shaping

Let

$$\tau_g = \min\{n \geq 1 : \text{the } n\text{-th chosen instrument is } S \text{ and reports } g\}$$

be the number of interventions up to and including the successful SIC use. Every chosen action has immediate cost

$$c_t = 1. \tag{16}$$

Equivalently, every action has reward $r_t = -1$. There is no extra reward for applying the “correct” Pauli, no reward for decreasing a supplied coordinate distance, and no terminal bonus. The only special event is that the successful SIC branch ends the episode and therefore eliminates all future costs. With no discounting, the return is

$$G_0 = \sum_{t=0}^{\tau_g-1} r_t = -\tau_g. \tag{17}$$

Maximizing expected return is exactly the same problem as minimizing the expected number of physical instrument uses from an observed history:

$$J_g(h) = \inf_{\pi} \mathbb{E}_h^{\pi}[\tau_g]. \tag{18}$$

The unit cost is a modeling choice, albeit a transparent one. If laboratory time, energy, or information acquisition differs by instrument, replace $c_t = 1$ by action-dependent costs $c_a > 0$. The Bellman proof and the resulting geometry must then be recomputed. In particular, calling the SIC measurement “free” would change the geometry.

7.4 The exact Bellman equation really lives on histories

Without any compression, the first-action recursion is

$$J_g(h) = \min_{a \in \mathcal{A}} \left[1 + \sum_{o: hao \text{ nonterminal}} P(o \mid h, a) J_g(hao) \right].$$

This equation is conceptually exact and uses only action–outcome strings. It appears to require an infinite tree, because every different h is a different argument. The reason we can solve it finitely is not that the agent is handed a coordinate. It is that the quantum instrument itself has only four distinct *predictive* kinds of reachable history.

For any history h , define its experimentally meaningful next-SIC response vector

$$\mathbf{q}(h) = (P(\omega_1 \mid h, S), \dots, P(\omega_4 \mid h, S)).$$

An experimenter estimates this vector by reproducing the same preparation procedure h many times and appending the common SIC test. It contains no position coordinate. More generally, call two histories *causally equivalent*, written $h \sim h'$, when every possible future adaptive action–outcome experiment has the same distribution after h and h' .

Proposition 7.1 (The finite predictive quotient). *After an observed SIC outcome, every reachable raw history belongs to exactly one of four causal equivalence classes. Moreover,*

$$h \sim h' \iff \mathbf{q}(h) = \mathbf{q}(h').$$

The four possible response vectors have one component equal to 1/2 and the other three equal to 1/6.

Proof. This is an analyst’s proof from observable probabilities, not an update rule given to the agent. Conditional on a raw history h , let ρ_h denote the normalized quantum state obtained by multiplying the Kraus operators recorded in that history. The next-SIC probabilities are

$$q_o(h) = \frac{1}{2} \text{Tr}(\Pi_o \rho_h).$$

The four tetrahedral SIC effects $\Pi_o/2$ are linearly independent in the four-dimensional real vector space of Hermitian qubit operators. Therefore $\mathbf{q}(h)$ determines ρ_h uniquely. Equal response vectors imply equal conditional density matrices, which in turn imply equal probabilities for every future sequence of instruments and outcomes. Conversely, causal equivalence includes the next-SIC experiment, so it implies equal response vectors.

Immediately after SIC outcome o , (13) gives $\rho_h = \Pi_o$. Each of the four instruments preserves the set $\{\Pi_o : o \in \Omega\}$ on every observed branch: X and Z permute it; success of $\tilde{Y}$ permutes it while failure leaves it unchanged; and a SIC outcome prepares one member of it. Induction on the length of the raw history proves that there are no other reachable conditional states. Equation (12) gives the stated four response vectors, which are distinct because their 1/2 peaks occur at different observed tokens. $\square$

The density matrix appears only in this proof that the observable relation is finite. Neither ρ_h nor a binary coordinate is an observation supplied to the controller. An agent may retain the full string h ; the theorem says that histories with the same future-test statistics have the same optimal continuation value. If $h \sim h'$, then every policy tree available after h has an identically distributed copy after h' , and hence

$$J_g(h) = J_g(h').$$

We may consequently write $[h]$ for a predictive class and solve on the four classes without assuming that the agent ever represents or updates that class online.

7.5 The undiscounted Bellman equation

The hodological construction uses the stochastic-shortest-path convention

$$\boxed{\gamma = 1.} \quad (19)$$

For any nonterminal predictive history class $\xi = [h]$, define

$$Q_g(\xi, a) = 1 + \sum_{o: hao \text{ nonterminal}} P(o \mid h, a) V_g([hao]). \quad (20)$$

This is well defined because causally equivalent histories have identical future distributions. The terminal transition has no continuation term. The optimality equation is

$$\boxed{V_g(\xi) = \min_{a \in \mathcal{A}} Q_g(\xi, a).} \quad (21)$$

Using $\gamma = 1$ is legitimate here even though many introductory RL algorithms emphasize $\gamma < 1$. This is an episodic proper stochastic-shortest-path problem: the policy “repeat the SIC” terminates with positive probability on each attempt, all costs are positive, and the optimal expected stopping time is finite. Discounting is therefore not needed to make the objective well defined. More importantly, expected hitting time is the quantity being interpreted as spatial distance.

Equation (21) does not itself magically produce the numbers. The calculation has two steps: evaluate a candidate history policy by solving its linear recurrence, and then insert the resulting values into *every* action expression to certify that none is smaller. The next section performs both steps. The verification covers arbitrary history-dependent policies, because any extra dependence on the past can only choose or randomize among the same four actions from a predictive class; it cannot alter their future outcome laws.

8 Full exact history-quotient analysis at $\gamma = 1$

Fix one observed SIC token g as the goal. Do not assign coordinates to the other tokens. For a raw history h , introduce the observable continuation probabilities

$$\begin{aligned} r_g(h) &:= P(g \mid h, S), \\ r_g^X(h) &:= P(g \mid h, X, \bullet_X, S), \\ r_g^Z(h) &:= P(g \mid h, Z, \bullet_Z, S), \\ r_g^{Ys}(h) &:= P(g \mid h, \tilde{Y}, s, S). \end{aligned}$$

For example, $r_g^X(h)$ is measured by reproducing the preparation history h , pressing the opaque button called X , observing its acknowledgement, then applying the SIC and recording how often token g occurs. It does not mean that the agent calculates a new position after pressing X .

Definition 8.1 (Predictive-class terminology relative to a goal). Among the four reachable causal classes of [Theorem 7.1](#), define

$$\begin{aligned} G &:= \{[h] : r_g(h) = 1/2\}, \\ E_X &:= \{[h] : r_g(h) = 1/6, r_g^X(h) = 1/2, r_g^Z(h) = 1/6\}, \\ E_Z &:= \{[h] : r_g(h) = 1/6, r_g^Z(h) = 1/2, r_g^X(h) = 1/6\}, \\ D &:= \{[h] : r_g(h) = r_g^X(h) = r_g^Z(h) = 1/6, r_g^{Ys}(h) = 1/2\}. \end{aligned}$$

The class G is the *goal-response class*. The classes E_X and E_Z are called the two *edge classes*: operationally, an edge class means exactly “one empirically identified deterministic intervention away from the goal-response profile.” In E_X , X is the *selected deterministic button*; in E_Z , it is Z . The symbol D denotes the remaining or *retry class* at this stage.

The words “edge” and, later, “diagonal” are retrospective geometric names. They are justified only after the excess costs are found to be 1 and $\sqrt{2}$. They are not assumptions used to obtain those costs. All four definitions concern distributions of future observed outcomes. An agent can learn them from repeated history–test experiments without being told a square, an origin, a bit string, or an action-permutation table. The symbols G, E_X, E_Z, D are analyst names for equivalence classes of whole histories; they are not a location register maintained during a trajectory.

The empirically testable class-transition facts are:

history class	deterministic buttons	$\tilde{Y}$ branches	next SIC outcomes
G	each leads to an edge class	$f : G, s : D$	terminal $g : 1/2$, edges total $1/3$, $D : 1/6$
$E_i, i \in \{X, Z\}$	button i leads to G , the other to D	both branches are edge classes	terminal $g : 1/6$, edges total $2/3$, $D : 1/6$
D	each leads to an edge class	$f : D, s : G$	terminal $g : 1/6$, edges total $1/3$, $D : 1/2$

Here “leads to class C ” means that appending the displayed action and outcome produces a longer history causally equivalent to histories in C . It is a statement about continuation distributions, not an instruction to update a hidden label. For example, the history-only policy can literally use the rules

- the raw history belongs to G : choose SIC;
- the raw history belongs to E_i : choose the empirically selected deterministic button, then SIC;
- the raw history belongs to D : choose $\tilde{Y}$ until outcome s , then SIC.

After a non-goal SIC result, the observed new token begins a new rule. Between the button press and the SIC test, the policy conditions on the literal acknowledgement or success/failure string; it never asserts “my position is now such-and-such.”

8.1 Step 1: evaluate the candidate history policy

By symmetry the two edge classes have the same value. Let v_0, v_1, v_2 be the expected remaining intervention counts of the displayed history policy from classes G, E_i, D , respectively. These are not guessed distances; they are unknowns in three elementary waiting-time recurrences.

From an edge history, the selected deterministic button costs one intervention. The policy then has the same future outcome statistics as a goal-profile history, so

$$v_1 = 1 + v_0.$$

From a D -history, one $\tilde{Y}$ attempt costs one. Failure, of probability $1 - p$, gives a causally equivalent D -history; success, of probability $p = 1/\sqrt{2}$, gives the goal response profile. Hence

$$v_2 = 1 + (1 - p)v_2 + pv_0, \quad \text{so} \quad v_2 = v_0 + \frac{1}{p} = v_0 + \sqrt{2}.$$

Finally, from a G -history the policy chooses SIC and pays one. Outcome g , of probability $1/2$, terminates and contributes no continuation. The two edge tokens have total probability $1/3$, and the remaining token has probability $1/6$. Therefore

$$v_0 = 1 + \frac{1}{3}v_1 + \frac{1}{6}v_2.$$

Substitution gives

$$\begin{aligned} v_0 &= 1 + \frac{1}{3}(v_0 + 1) + \frac{1}{6}(v_0 + \sqrt{2}), \\ \frac{1}{2}v_0 &= \frac{4}{3} + \frac{\sqrt{2}}{6}, \\ v_0 &= \frac{8 + \sqrt{2}}{3}. \end{aligned}$$

Thus policy evaluation, using only the observable transition frequencies in the preceding table, yields

$$c = \frac{8 + \sqrt{2}}{3}, \quad v_0 = c, \quad v_1 = c + 1, \quad v_2 = c + \sqrt{2}.$$

8.2 Step 2: verify optimality against every first action

Solving the recurrences evaluates one candidate policy; it does not yet prove optimality. We now compute the cost of every possible first instrument, followed by the candidate continuation. If their minimum reproduces (v_0, v_1, v_2) , the Bellman verification theorem proves global optimality. The following table evaluates *every* action, not only the optimal one. At an edge, “selected deterministic” means whichever opaque button repeated future-SIC tests show to lead to the goal response profile; the other is “unselected.”

predictive class	action	exact Q -value	optimal?
G	either deterministic button	$c + 2$	no
G	$\tilde{Y}$	$c + 2$	no
G	SIC	c	yes
E_i	selected deterministic button	$c + 1$	yes
E_i	unselected deterministic button	$c + 1 + \sqrt{2}$	no
E_i	$\tilde{Y}$	$c + 2$	no
E_i	SIC	$c + 1 + (2 + \sqrt{2})/9$	no
D	either deterministic button	$c + 2$	no
D	$\tilde{Y}$	$c + \sqrt{2}$	yes
D	SIC	$c + \sqrt{2} + (8 - 5\sqrt{2})/9$	no

Here are the calculations behind the table. From class G , either deterministic button has edge-profile continuation, so its value is $1 + v_1 = c + 2$. Random Y fails back to an equivalent G -history with probability $1 - p$ and succeeds to class D with probability p , whence

$$Q_{\tilde{Y}} = 1 + (1 - p)v_0 + pv_2 = 1 + c + p\sqrt{2} = c + 2.$$

The SIC terminates with probability $1/2$, produces an edge-profile history with total probability $1/3$, and a D -profile history with probability $1/6$:

$$Q_S = 1 + \frac{1}{3}v_1 + \frac{1}{6}v_2 = c.$$

From E_i , the empirically selected deterministic button incurs one step and has G -profile continuation, giving $1 + v_0 = v_1$. The unselected button has D -profile continuation. Random Y has an edge-profile continuation on either branch, so its continuation value is always v_1 . A SIC use

terminates with probability $1/6$, has an edge continuation with total probability $2/3$, and a diagonal continuation with probability $1/6$. Therefore

$$\begin{aligned} Q_S(\text{edge}) &= 1 + \frac{2}{3}v_1 + \frac{1}{6}v_2 \\ &= v_1 + \frac{2 + \sqrt{2}}{9} > v_1. \end{aligned}$$

From D , either deterministic button has edge-profile continuation and costs $1 + v_1 = c + 2$. Random Y has G -profile continuation upon success and an equivalent D -history upon failure:

$$\begin{aligned} Q_{\tilde{Y}}(\text{diagonal}) &= 1 + (1 - p)v_2 + pv_0 \\ &= v_2, \end{aligned}$$

where the last equality is exactly the geometric-retry identity $p(v_2 - v_0) = p\sqrt{2} = 1$. Finally,

$$\begin{aligned} Q_S(\text{diagonal}) &= 1 + \frac{1}{3}v_1 + \frac{1}{2}v_2 \\ &= v_2 + \frac{8 - 5\sqrt{2}}{9} > v_2. \end{aligned}$$

Both strict gaps are positive. The complete optimal rule on history classes is consequently

$$\pi_g^*([h]) = \begin{cases} S, & [h] = G, \\ \text{the empirically selected deterministic button}, & [h] = E_i, \\ \tilde{Y}, & [h] = D. \end{cases} \quad (22)$$

Numerically, $c \simeq 3.138071$. The self, edge, and diagonal values are, respectively, 3.138071, 4.138071, and 4.552285. Subtracting the unavoidable goal-reporting cost v_0 gives $0, 1, \sqrt{2}$, the desired square radii.

8.3 Why this proves optimality over all histories

The table shows that the candidate vector $V = (v_0, v_1, v_1, v_2)$ satisfies the minimum equation (21); the displayed history policy attains equality. For any other possibly randomized, time-dependent, or full-history policy, the Bellman inequalities give at every nonterminal time

$$V([h_t]) \leq 1 + \mathbb{E}[V([h_{t+1}]) \mid h_t, a_t].$$

Iterating to the stopping time shows that its expected accumulated unit cost is at least $V([h_0])$. A policy that fails to terminate with positive probability has infinite undiscounted cost and cannot improve the result. The candidate policy is proper and all its displayed recurrences are equalities, so its expected cost is exactly $V([h_0])$. This is the Bellman verification argument.

This proof quantifies over every raw-history policy without listing its branches. Histories are omitted only after [Theorem 7.1](#) proves that they have identical future statistics. A history-dependent controller can use its past as an internal clock or random seed, but from a fixed predictive class that merely selects a mixture of the four first-action costs in the table; a mixture cannot be less than their minimum.

9 What discounting would change

For $0 \leq \gamma < 1$, the discounted step-cost objective would instead be

$$V_g^{(\gamma)}(\xi) = \inf_{\pi} \mathbb{E}_{\xi}^{\pi} \left[\sum_{t=0}^{\tau_g-1} \gamma^t \right] = \inf_{\pi} \frac{1 - \mathbb{E}_{\xi}^{\pi}[\gamma^{\tau_g}]}{1 - \gamma}. \quad (23)$$

It has Bellman equation

$$V_g^{(\gamma)}(\xi) = \min_a \left\{ 1 + \gamma \sum_{o: hao \text{ nonterminal}} P(o | h, a) V_g^{(\gamma)}([hao]) \right\}, \quad \xi = [h]. \quad (24)$$

For a deterministic stopping time, (23) is a monotone transform of the number of steps. For a random stopping time it depends on the entire distribution of τ_g , not only its mean. It preferentially weights chances of very early completion. A common alternative reward, zero on ordinary transitions and +1 on the terminal SIC outcome, maximizes $\mathbb{E}[\gamma^{\tau_g-1}]$. For $0 < \gamma < 1$ this selects the same policies as minimizing (23), but neither objective is the expected hitting time that defines the present hodological metric.

The distinction can be solved exactly on the four predictive history classes. Again let v_0, v_1, v_2 denote the G , edge-profile, and D -profile discounted costs. There are three policy regimes. Define

$$\gamma_* = \frac{3 + 3\sqrt{2} - \sqrt{15 + 12\sqrt{2}}}{2} \simeq 0.7941944732. \quad (25)$$

At $\gamma = 0$, every action has the same one-step discounted cost, so every first action ties. For $0 < \gamma \leq 1/2$, a SIC use is optimal in all three shells. The following exact solution also extends continuously to $\gamma = 0$:

$$v_0 = \frac{6 - 2\gamma}{6 - 5\gamma}, \quad v_1 = v_2 = \frac{6}{6 - 5\gamma}. \quad (26)$$

The equality $v_1 = v_2$ shows that strong discounting erases the distinction between an edge and the diagonal. To derive these expressions, observe that from any non-goal response profile a SIC reports g with probability $1/6$; under the symmetric all-SIC solution its total nonterminal continuation is $(5/6)v_1$. In class G , the nonterminal probability is $1/2$.

For $1/2 \leq \gamma \leq \gamma_*$, SIC is optimal at self, an axial move is optimal at an edge, and SIC remains optimal at the diagonal. With

$$\Delta = 2\gamma^3 - 6\gamma^2 - 9\gamma + 18,$$

the exact values are

$$v_0 = \frac{2(9 - \gamma^2)}{\Delta}, \quad v_1 = \frac{3(-2\gamma^2 + 3\gamma + 6)}{\Delta}, \quad v_2 = \frac{6(\gamma + 3)}{\Delta}. \quad (27)$$

They follow by solving the three linear policy-evaluation equations

$$v_0 = 1 + \gamma \left(\frac{1}{3}v_1 + \frac{1}{6}v_2 \right), \quad v_1 = 1 + \gamma v_0, \quad v_2 = 1 + \gamma \left(\frac{1}{3}v_1 + \frac{1}{2}v_2 \right).$$

For $\gamma_* \leq \gamma \leq 1$, the undiscounted policy structure is restored: SIC at self, axial movement at an edge, and retry- Y at the diagonal. Put

$$a_{\gamma} = 1 - \gamma \left(1 - \frac{1}{\sqrt{2}} \right).$$

Then an economical exact form is

$$v_0 = \frac{1 + \gamma/3 + \gamma/(6a_\gamma)}{1 - \gamma^2/3 - \gamma^2/(6\sqrt{2}a_\gamma)}, \quad (28)$$

$$v_1 = 1 + \gamma v_0, \quad v_2 = \frac{1 + \gamma v_0/\sqrt{2}}{a_\gamma}. \quad (29)$$

The first transition occurs when the edge is indifferent between reporting and moving, which reduces exactly to $\gamma = 1/2$. The second occurs when the diagonal is indifferent between reporting and retrying Y . Substituting (27) gives

$$\gamma^2 - 3(1 + \sqrt{2})\gamma + 3 + \frac{3\sqrt{2}}{2} = 0,$$

whose only root in $[0, 1]$ is (25). Direct substitution checks that all unselected actions have no smaller Q -value in their stated intervals. At a boundary the adjacent policies tie.

At $\gamma = 1$, (28)–(29) reduce to

$$(v_0, v_1, v_2) = (c, c + 1, c + \sqrt{2}).$$

Thus the Euclidean square is not invariant under arbitrary discounting. It is specifically a geometry of expected undiscounted intervention count. The computed phase diagram in Fig. 1 shows the approach to that limit.

10 A compact derivation of the SIC reporting baseline

For reference, we restate the common baseline in compact predictive-class form. As specified in Section 7, goal g means: perform the SIC action and observe outcome token g . Every movement and every SIC report costs one intervention. If a SIC outcome is not g , the episode continues from a raw history ending in the newly observed token. The finite equations below are licensed by Theorem 7.1, not by supplying the agent a location variable.

By response symmetry, write v_0, v_1, v_2 for optimal cost in predictive classes G, E_i, D , respectively. The proposed history rule reports in G , uses the empirically selected deterministic button in E_i , and retries $\tilde{Y}$ in D . The two movement rules give

$$v_1 = 1 + v_0, \quad v_2 = \sqrt{2} + v_0. \quad (30)$$

In class G , the SIC reports g with probability $1/2$, terminating the episode. Its two edge outcomes have total probability $1/3$, and its diagonal outcome has probability $1/6$. Consequently

$$v_0 = 1 + \frac{1}{3}v_1 + \frac{1}{6}v_2 \quad (31)$$

$$= 1 + \frac{1}{3}(v_0 + 1) + \frac{1}{6}(v_0 + \sqrt{2}), \quad (32)$$

so

$$\boxed{v_0 = c := \frac{8 + \sqrt{2}}{3}}. \quad (33)$$

It follows that the full reported-goal value matrix is

$$\boxed{V_g(\omega) = c + D(\omega, g), \quad \omega \in \Omega.} \quad (34)$$

The remaining Bellman inequalities confirm optimality. In G , choosing a movement instrument first costs $c + 2 > c$. In E_i , reporting immediately costs

$$1 + \frac{2}{3}v_1 + \frac{1}{6}v_2 > v_1,$$

while the selected deterministic button attains v_1 . In D , reporting costs

$$1 + \frac{1}{3}v_1 + \frac{1}{2}v_2 > v_2,$$

and a two-deterministic-button route costs $c + 2 > c + \sqrt{2}$; retry attains v_2 . Thus (34) is the optimal Bellman solution, not merely the value of a selected policy.

The nonzero repeated-token value $V_g(g) = c$ is the common overhead of requesting a stochastic SIC report. Hodological separation is obtained by subtracting this goal-specific self baseline:

$$V_g(\omega) - V_g(g) = D(\omega, g). \quad (35)$$

The exact Euclidean square therefore survives when places and goals are both defined entirely by experienced SIC outcomes.

Before the first SIC action, an agent with an unknown preparation occupies an epistemic initial condition. Its first SIC outcome supplies the first token in the recorded history. Subsequent control uses only that token, chosen actions, and observed acknowledgements or success/failure events; the policy neither receives nor updates a concealed binary-coordinate variable.

11 Exact computation and finite-sample simulation

The companion program `sic_anchored_square.py` constructs all density matrices and Kraus operators, solves the four stochastic-shortest-path problems, runs Monte Carlo navigation, and learns randomly renamed Pauli permutations from anchor–action–SIC experiments. Nine focused tests verify Kraus completeness, branch preparation, the $1/2, 1/6$ response kernel, the analytic Bellman solution, every undiscounted action value, the three discounted policy regimes, the raw-history predictive quotient, and optimal-policy structure.

At seed 20260813, exact value iteration gives

$$\max_{x,g} |V_g(x) - c - D(x, g)| = 2.58 \times 10^{-14},$$

and baseline subtraction recovers D to 9.10×10^{-15} . Across 50,000 Monte Carlo episodes for each of the 16 ordered pairs, maximum mean-cost error is 0.02831. In 500 independent opaque-label trials per sample size, the complete four-action response permutations are recovered in every trial at 100 samples per anchor/action pair. Across a 503-point discount scan, the closed forms in [Section 9](#) agree with numerical Bellman iteration to 3.02×10^{-14} .

As a direct audit of [Theorem 7.1](#), the program enumerates every action–outcome extension through depth five without propagating a location label. There are

$$4 \times 8^5 = 131,072$$

distinct raw histories at that depth: four choices of initial observed SIC token and eight possible chosen-action/outcome pairs per extension ($4 + 1 + 2 + 1$). Computing the distribution of a future SIC test from each history yields exactly four distinct response vectors, with maximum deviation from the analytic $(1/2, 1/6, 1/6, 1/6)$ permutations below numerical precision. The table is saved as `history_predictive_quotient.csv`. This enumeration is a check of the quotient theorem, not the method used to solve the Bellman equations.

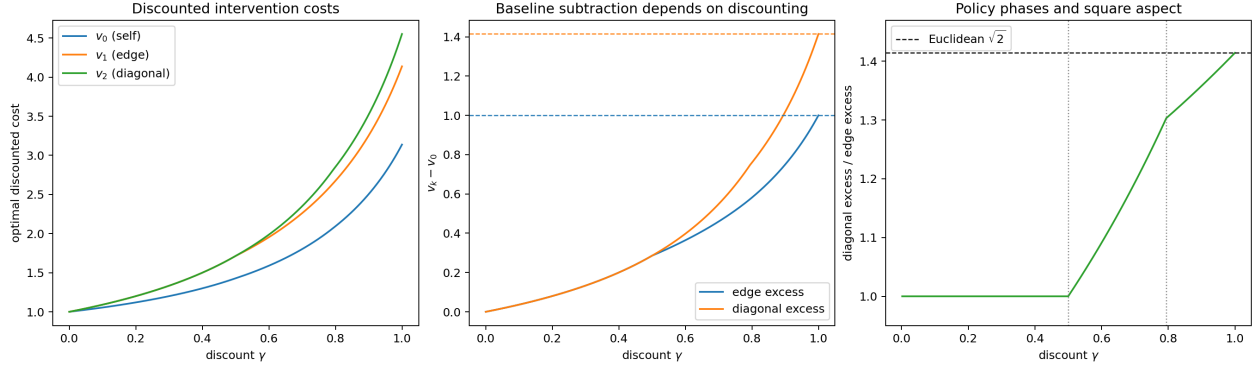


Figure 1: **Discounting changes both value and geometry.** Left: exact discounted self, edge, and diagonal intervention costs. Center: after subtracting the self reporting baseline, the edge and diagonal separations reach 1 and $\sqrt{2}$ only at $\gamma = 1$. Right: the diagonal-to-edge ratio and the two exact policy thresholds $\gamma = 1/2$ and $\gamma = \gamma_*$. Strong discounting makes all non-goal SIC-report policies optimal and collapses edge and diagonal into one shell.

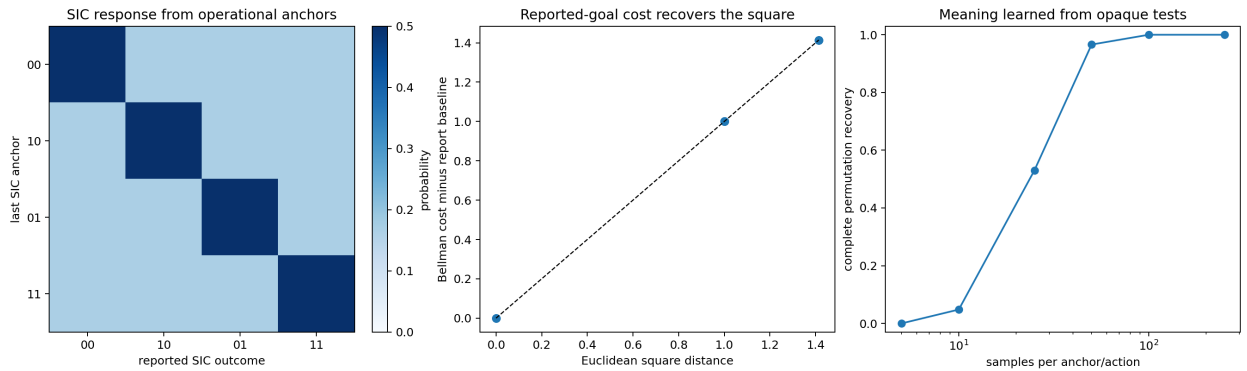


Figure 2: **The square grounded in SIC outcomes.** Left: after an observed SIC anchor, the same outcome has probability $1/2$, every other outcome has probability $1/6$, and every branch prepares the state named by its token. Center: subtracting the common reporting baseline from exact Bellman costs recovers side 1 and diagonal $\sqrt{2}$. Right: randomly renamed Pauli response permutations are learned from future SIC tests without exposing binary coordinates.

12 Why D is exactly a two-dimensional Euclidean metric

The entries of D already match the familiar side and diagonal lengths of a unit square. Schoenberg's test certifies this without presupposing the coordinates in (9).

First square every entry:

$$D^{\circ 2} = \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{pmatrix}. \quad (36)$$

Every row mean, every column mean, and the grand mean of this matrix equal one. With

$$J = I - \frac{1}{4}\mathbf{1}\mathbf{1}^T,$$

the double-centered Gram matrix is

$$B = -\frac{1}{2}JD^{\circ 2}J \quad (37)$$

$$= \begin{pmatrix} \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}. \quad (38)$$

To see the geometry directly, center the square coordinates:

$$X = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \quad (39)$$

Multiplication gives $XX^T = B$. Therefore B is positive semidefinite. The two columns of X are orthonormal, so the eigenvalues of B are

$$1, 1, 0, 0.$$

It has rank two, and classical multidimensional scaling reconstructs the four rows of X , up to rotation and reflection. This proves intrinsically that the hodological metric (15) is an exact two-dimensional Euclidean metric.

13 What the theorem proves—and what it assumes

The theorem proves three useful facts.

1. The Kraus operators form valid quantum instruments.
2. The desired metric is achieved by a concrete policy.
3. The triangle inequality is a global certificate against every adaptive shortcut.

It does not derive the metric from an otherwise neutral quantum dynamics. The desired length is inserted into the success law $p_h = 1/d_G(\mathbf{e}, h)$. In the square, the Euclidean diagonal is obtained because the diagonal action was assigned success probability $1/\sqrt{2}$. A different probability would produce a different hodological diagonal.

It also does not imply that instrument outcomes localize the agent. In a random-unitary retry action, success/failure reports whether a chosen translation occurred; it need not reveal the starting state. Finally, the four qubit states form a tetrahedron in Bloch-state geometry even though their optimal costs form a square. This separation is intentional: hodological distance measures difficulty of controlled goal achievement, not quantum state overlap.

The SIC extension in [Sections 5.3, 7 and 10](#) repairs the location-token problem in a precise, history-only sense: place goals are future SIC outcomes, and the controller is given only its action and outcome record. It does not retrospectively identify the unknown premeasurement state, nor make a deterministic movement acknowledgement location-informative. The exact finite calculation is possible because future SIC response distributions induce four causal equivalence classes of raw histories. Those classes are derived from observable consequences; they are not coordinates supplied to or updated by the agent.

14 A compact checklist for using the theorem

Before applying the retry theorem to a new construction, check:

1. Is every desired displacement represented by a valid group element?
2. Is the orbit faithful after quotienting global phase and the fiducial stabilizer?
3. Does the side of multiplication match the side of metric invariance?
4. Has the metric been normalized so every $p_h \leq 1$?
5. Does failure truly leave the operational coordinate unchanged?
6. Is the goal defined by the physical orbit, a predictive equivalence class, or an external group-product monitor? These are not interchangeable.
7. Is $p_h = 1/d_G(\mathbf{e}, h)$ physically motivated, learned, or simply supplied by design?

For Eq. (43), all algebraic conditions hold exactly. The remaining scientific limitation is the last one: the Euclidean success law is engineered rather than derived.